Saumya Mundra

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Research Interests

Geometric computer vision, 3D/4D perception, visual reasoning, multimodal learning, and robotic vision.

Education

International Institute of Information Technology, Hyderabad Jan 2026 – Jul 2027 (expected)
MS by Research in Computer Science and Engineering **CGPA: 9.33**
Relevant coursework: Computer Vision, Statistical Methods in AI (SMAI), Responsible and Safe AI (RSAI), Mobile Robotics.

Indian Institute of Technology, Hyderabad 2020 – 2024
B.Tech. in Chemical Engineering **CGPA: 8.69**
Relevant coursework: Deep Learning, Probability, Optimization, Data Structures, Applied Mathematics.

Research Experience

CVIT, IIIT Hyderabad Jan 2026 – Present
MS Research Student
Advisors: Prof. C. V. Jawahar and Prof. Vineeth N. Balasubramanian; collaborator: Dr. Ajoy Mondal.

- **Geometric reasoning in MLLMs:** Formulated controlled diagnostics for visual correspondence and **2D transformations** (translation, rotation, scaling, and compositions), probing transformation type, direction, and magnitude.
- Evaluated zero-shot reasoning and LoRA adaptation to study the learnability of geometric transformations from visual inputs. *Manuscript in preparation.*

CVIT, IIIT Hyderabad Aug 2024 – Jan 2026
Research Fellow

- **MIMO-Bench – multi-instance, multi-object visual reasoning:** Co-developed a **3,802-question** benchmark for attribute binding and numerical counting with visually similar distractors and compositional referring expressions; coordinated annotation, human answering, and verification.
- Designed **group-level accuracy** and global/local counting metrics; isolated the effects of referring-expression complexity and distractors through visual prompting, expression simplification, graying, and inpainting.
- On real-image attribute binding, the strongest MLLM achieved **52.8%** accuracy versus **89.3%** for humans; visual prompting improved model accuracy by **15.5 percentage points** on average. Relevant-region cropping helped, while chain-of-thought prompting did not reliably improve results. *arXiv preprint; under review.*
- **MIST – multilingual incidental scene text (WACV 2026):** Defined quantitative measures of **incidentalness** using text density and relative text area; showed that detection deteriorates as text becomes more incidental, even on existing benchmarks.
- Constructed **12,000 images** with approximately **576,000 word annotations across 11 scripts**; developed a semi-automatic annotation pipeline, piloted polygon refinement versus manual annotation, and independently coordinated **20+ annotators** and systematic quality control.
- Benchmarked four scene-text detectors (best **62.7% F1**; **52.8%** on the most incidental subset); demonstrated cross-dataset generalization and improvements from MIST pretraining under limited target-domain training data.

Publications and Preprints

Peer-reviewed conference papers

1. **Saumya Mundra**, Ajoy Mondal, C. V. Jawahar. "MIST: Multilingual Incidental Dataset for Scene Text Detection." *IEEE/CVF WACV*, 2026, pp. 7658–7667. [paper] [dataset]
2. Ajoy Mondal, **Saumya Mundra**, Avijit Dasgupta, C. V. Jawahar. "UniTabBank: A Large Scale Multi-Lingual, Multi-Layout, Multi-Type, Multi-Format Dataset for Table Detection." *IEEE/CVF WACV*, 2026, pp. 6486–6495. [paper]
3. Nagasree Keerthi Pujari, **Saumya Mundra**, Kishalay Mitra. "Accurate Modelling of Wind Characteristics Using Temporal Convolutional Neural Networks." *11th Indian Control Conference (ICC)*, 2025, pp. 624–629. [doi]

Book chapter

4. Nagasree Keerthi Pujari, **Saumya Mundra**, Yash Sharma, Kishalay Mitra. "Wind Behavior Modeling Towards Improvement of Sustainable Energy Generation: Generative Adversarial Networks versus Diffusion Models." In *Data-Driven Insights and Analytics for Measurable Sustainable Development Goals*, Elsevier, 2025, pp. 147–161. [doi]

Preprint

5. **Saumya Mundra**, Ajoy Mondal, Vineeth N. Balasubramanian, C. V. Jawahar. "What do MLLMs know about 'Waldo'? Evaluating Visual Reasoning under Multi-Instance Multi-Object Challenges." *arXiv:2610.38957*, 2026. [arXiv]

Selected Research Projects

Geometry for Robot Navigation *Collaboration with Prof. Sourav Garg, 2026 – Present*

- Investigating whether **persistent geometric state** learned during mapping can replace repeated visual-place retrieval and multi-view processing for goal-conditioned robot navigation, using streaming 3D reconstruction models including CUT3R, TTT3R, and Plann3r.
- Exploring state updates across image windows and single-query costmap prediction to reduce repeated geometric computation. *Ongoing research.*

RADBench-AD: Counterfactual Referring Expressions in Driving
Responsible and Safe AI Systems course project, 2026

- Formulated **referent-switching** and **invalid-reference rejection** tasks for driving scenes; devised a paired consistency score and led experimental design, implementation, evaluation, and analysis.
- Found that general-purpose MLLMs handled object switching more reliably than rejecting absent targets, while the driving-specific **UniDriveVLA** showed the opposite failure pattern.

Compositional Referring Expression Validity *Computer Vision course project, 2026*

- Extended LabelMe for fine-grained scene-graph annotations; constructed image-grounded valid expressions and hard negatives by perturbing **objects, attributes, and relations**.
- Compared multiple-choice with independent positive/negative judgments using **strict paired accuracy**; the best tested model achieved only **16.6%** on hard-negative pairs, exposing failures masked by multiple-choice evaluation.

Additional Research and Coursework

- **Incremental SfM (ongoing):** Developing an incremental reconstruction pipeline with triangulation, PnP-RANSAC, and bundle adjustment; planned comparison against COLMAP.
- **Wind modeling (IIT Hyderabad, 2023–24):** Formulated autoregressive wind-frequency map prediction; compared TCNs, LSTMs, and Transformers, applied Bayesian optimization, and explored GAN/diffusion synthesis (ICC 2025 paper; Elsevier chapter).

Research Presentations

Oral Presentation: "MIST: Multilingual Incidental Dataset for Scene Text Detection," *Indian Language OCR Workshop, ICVGIP 2025*, IIT Mandi, December 2025.

Teaching and Academic Service

- **Instructor, CVIT Summer Schools (2025, 2026):** Independently delivered "Introduction to Deep Learning through Code and Concepts," combining conceptual explanations with code demonstrations for **250+ participants**.
- **Student point of contact and session moderator, CVIT:** Helped organize visiting-researcher sessions, coordinated with speakers, introduced talks, moderated discussions, and facilitated Q&A, including sessions with **Dr. Vishnu Lokhande** (University at Buffalo), **Prof. Manmohan Chandraker** (UC San Diego), and **Prof. Pankaj Pansari** (Plaksha University).

Industry Experience

Unstudio.ai – Machine Learning Intern Jan – Mar 2024

- Developed Python-based data pipelines; supported SDXL/LCM-LoRA/ControlNet training and data debugging; explored language-model fine-tuning and inference deployment.

Cooler Screens – Machine Learning Intern May – Jul 2023

- Investigated label noise in large-scale image classification using confident learning, feature mixup, and area-under-margin analyses.

Fellowship.ai – Machine Learning Intern Oct – Dec 2022

- Implemented components of a portrait-relighting pipeline, including semantic-guided human matting, object placement, and image harmonization.

Leadership, Skills and Activities

Leadership: Student Placement Head (2023–24) and **Internship Head** (2022–23), IIT Hyderabad. Led student coordinator teams for campus recruitment and internships; organized *Synergy*, connecting students with executives from Visa and Cisco.

Programming and tools: Python, PyTorch, TensorFlow, Hugging Face, NumPy, scikit-learn, pandas, Matplotlib, Jupyter; working knowledge of C/C++, Bash, SQL.

Training: Deep Learning Specialization, DeepLearning.AI (2022).

Activities: Represented IIT Hyderabad in the Inter-IIT Sports Meet (football).